Extending the Detection Range for Low-Channel Roadside LiDAR by Static Background Construction

Hui Liu, Ciyun Lin, Bowen Gong, and Dayong Wu

***Abstract*— Further detection will increase traffic safety by perceiving unexpected traffic incidents earlier, allowing more time to make decisions and implement these decisions. Therefore, this article attempts to extend the detection range by using a low-channel roadside light detection and range sensor (LiDAR) sensor due to its low price and widespread employment in the future. The proposed method contains two major parts: static background construction and traffic objects detection. For static background construction, successive point cloud frames data were used to cover the most LiDAR scanning horizontal–vertical angle and obtain background information ultimately. Furthermore, the constructed static background was optimized to reduce the missing of the far range points and the occurrence of noise points. For vehicle and road user detection, the density-based spatial clustering of applications with noise (DBSCAN) algorithm was used to identify the near-range traffic objects. For far-range traffic objects detection, the trajectories of objects were extracted based on their moving direction and distance, and a fast Fourier transform (FFT) algorithm was used to filter the noise point and identify the vehicle and road user points. The experimental results show that the static background construction with the proposed method is more effective and uses fewer point cloud data. The average maximum detection range can be up to 100, 100, and 85 m for vehicles, cyclists, and pedestrians, respectively. The average precision (AP) and recall are 95.71% and 90.62%, respectively, which are better than the previous studies.**

***Index Terms*— Extending detection range, far-range detection, roadside light detection and range sensor (LiDAR) data, static background construction, traffic perception.**

I. Introduction

SINCE the news of Tesla and Google automatic vehicles (AVs) failing to identify the road users and leading to crash accidents were reported [1], [2], there is a consensus that it is not enough to provide reliable traffic information just by the sensors in AVs [3]. Infrastructure-based sensors, also called roadside sensors or roadside units, are critical in providing complementary and beyond visual range (BVR) traffic information for AVs to ensure high dependability, integrality, and credibility of traffic perception [4], [5].

The light detection and range sensors (LiDARs), as new-fashioned active visual sensors, are widely installed in AVs and roadsides to perceive vehicles, road users, and other obstacles [6]–[8]. However, to the best of our knowledge, the valid detection range of LiDAR sensors, laser beams less than 32, is 30–50 m in identifying the surrounding road users and vehicles effectively [9], [10], which is far less than its built-in detection range. For example, the claimed maximum detection ranges of Velodyne VLP-16, RoboSense RS-16, and Velodyne VLP-32E are up to 100, 150, and 200 m, respectively.

In practice, especially in the field of traffic safety, it would like to pursue a further detection range, which can provide more time to respond to an approaching unexpected traffic incident [11]. In addition, it is challenging for traffic management agents to make delicacy management in crowded urban environments without perceiving further enough to cover all the queuing vehicles. However, in most metropolitans, the queuing length in the signalized intersection is always greater than 50 m during the peak hours [8]. We believe that extending the detection range of roadside sensors can enhance the traffic safety for connected vehicles and expand the application in traffic management and control.

To extend the detection range of the LiDAR sensor, Wang and Goldluecke [12] proposed a Sparse-PointNet to extend detection range objects for AVs from mobile LiDAR data and cameras jointly. Zhang *et al.* [13] extracted vehicle and pedestrian points from roadside LiDAR data by constructing and optimizing an adaptive background. To our best knowledge, this is the first study dedicated to extending the detection range for AVs and roadside traffic perception, respectively. However, current researches in extending the detection range still have some issues to be enhanced: 1) it required that there was no obstacle between the far-range object and the sensor; 2) the pedestrians as the most vulnerable road users were often ignored in the far range detection; and 3) the accuracy and robustness will reduce sharply in complex and high-density traffic environment [14].

To fill the gap, we attempted to construct a static background as a funnel to filter out background point data as precisely as possible and suitable for different levels of traffic conditions.

Manuscript received November 20, 2021; revised February 12, 2022; accepted February 26, 2022. Date of publication February 28, 2022; date of current version April 5, 2022. This work was supported in part by the Graduate Innovation Fund of Jilin University under Grant 101832020CX150, in part by the Scientific Research Project of the Education Department of Jilin Province under Grant JJKH20221020KJ, and in part by the National Natural Science Foundation of China under Grant 51408257. *(Corresponding author: Ciyun Lin.)*

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This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2022.3155634, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2022.3155634

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Then, we extracted the trajectories from the multi-frame point cloud. To extend the detection range and detect the far range objects accurately, we used fast Fourier transform (FFT) to classify object points and noise points in the far range which the point cloud is sparse.

The major contributions of this article can be considered as follows.

1) A novel static background construction method was proposed to filter the background point precisely and suit different levels of traffic conditions.
2) The FFT algorithm was first introduced in object points and noise points classification in the sparse point cloud scene.
3) The method of extending detection range with low-channel LiDAR work well in complex traffic flow environments. There can be different densities of traffic flow between the far range and the LiDAR sensor.
4) The vehicle and cyclist detection distance can extend to 100 m, equal to LiDAR's detection range. The pedestrian detection distance can extend to 85 m. The distance, average precision (AP), and average recall (AR) are further or higher than the previous studies in complex traffic environments.

The remainder of this article is organized as follows. Section II reviews related research on LiDAR-based and image-based object detection, background filtering, and detection extending. Section III introduces the used LiDAR sensor and its data. Section IV presents the method of static background construction. Section V describes the method of extending detection. In Section VI, experiments are conducted to illustrate the performance of our method. This article is concluded in Section VII with the summary and possible future work.

II. Related Work

To distinguish our work from the existing studies, previous researches on vehicle and road user identification using LiDAR point clouds is reviewed. We also review the literature for extending the detection range for the LiDAR and camera sensor to the best of our knowledge.

A. LiDAR-Based 3-D Vehicle and Road User Detection

LiDAR-based 3-D vehicle and road user detection methods can be divided into deep learning methods and filtering-based methods.

The deep learning methods take the point cloud as input directly to generate the 3-D boundary box of each target object. These methods can be classified into three main categories: point-based methods [15], [16], voxel-based methods [11], [17], and bird-eye-view (BEV)-based methods [18]. Point-based approaches usually apply point feature extractors, such as PointNet [16] and PointNet++ [15] directly on every single point in scenes with large-scale point clouds. Voxel-based methods typically discretize the points with ordered structures and then exploit 2-D or 3-D convolutional neural networks (CNNs) for feature extraction. The voxel-based methods proposed describe each voxel with a voxel feature encoding (VFE) layer. BEV-based methods can run at high speed but with reduced accuracy.

However, these methods should work in a supervised learning framework to achieve better performance, which requires considerable label data. In addition, the learning architecture is usually designed for high-channel LiDAR data. For example, 64-channel or 128-channel LiDAR has been used in the KITTI [19] and Waymo [20] datasets, leading to poor performance in low-channel LiDAR data for some CNN-based methods [21], [22].

The filtering-based detection methods are composed mainly of background filtering, clustering, or segmentation. The filtering process is the basis and foremost step, which can directly affect the final results and will be discussed in Section II-B.

B. LiDAR Data Background Filtering

For onboard LiDAR, Azim and Aycard [23] used an octree-based occupancy grid representation to model the dynamic environment around the vehicle and detect moving objects based on inconsistencies between scans. Asvadi *et al.* [24] built a voxel-grid model for static and moving obstacle detection by using discriminative analysis and ego-motion information. Dewan *et al.* [25] presented a novel model-free approach for detecting and tracking dynamic objects in 3-D LiDAR scans obtained by a moving sensor.

For mobile LiDAR, the surrounding environment changes dynamically, including trees, buildings, and traffic flow conditions. Unlike onboard LiDAR, background objects are stationary in the scenes while using the roadside LiDAR scanning. However, vehicles and road users are distributed randomly within the LiDAR scanning range. The feature vehicles or road users' points, such as density, shape, and mobility, will change dynamically, which is widely used in background filtering.

Wu *et al.* [9] proposed an automatic density-based background filtering algorithm (3D-DSF) using the density feature changing in the LiDAR scanning range. With the shape feature of the target's points, Song *et al.* [26] split all points into groups according to laser ID and then used the shape formed by different laser IDs of the LiDAR sensor to filter the background points. Zhang *et al.* [13] used the distance feature of the point cloud to construct the background dataset by transforming the 3-D point cloud data into 2-D images. We used mobility and point shapes to distinguish moving and stationary objects and filtered the background points in our previous studies; the average accuracy was up to 98.7% in the near range [10], [27].

For roadside LiDAR, the density of point cloud diminishes sharply with distance. The vehicles or road users' points will turn to very sparse. Even a vehicle or road user merely obtain a laser point in far range. That will result in insufficient information to identify and classify the laser point. Therefore, the accuracy of background filtering should be prompted in further object detection, especially in sparse point cloud.

C. Extending Detection Range for Visual Sensor

The accuracy and robustness of the far-range objects are still a critical problem for visual sensors, such as cameras and LiDARs. The essential reason is that the pivotal features

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of the object are hard to extract from the sparse points or pixels in the far range. Experts in machine vision proposed many models and algorithms to extend the detection range and improve the detection precision for visual sensor far range detection.

Some multi-scale methods were proposed recently for camera sensors to detect far-range vehicles and road users more accurately. SAF RCNN [28] and MS-CNN [29] extended Fast RCNN [30] and Faster RCNN [31] to integrate a near-scale and a far-range sub-network into a unified architecture. F-DNN [32] used multiple deep classifiers combined with soft filters to validate its proposal. However, these models or algorithms focused on using convolutional neural networks to extract the features of the near-range and far-range targets, which significantly increase the computational burden in far-range object detection [33]. Therefore, Li *et al.* [34] proposed a conditional random field (CRF)-based message passing mechanism to explore complementary information among multi-scale features entirely. Zhang *et al.* [35] proposed a key-point-guided super-resolution network, which could extract the information in key points of objects for detection and extended the max valid detection range to 20–50 pixels in height for image data [11].

For onboard LiDAR, Wang and Goldluecke [12] proposed a Sparse-PointNet, which can jointly detect objects from multimodal sensor data and improve the performance by increasing the accuracy of the far-range vehicle and road user features obtained. For roadside LiDAR, Zhang *et al.* [13] extended the detection range for the far-range vehicles by enriching background points. However, these methods will be limited in practical application scenarios, especially for intersections with heavy traffic. In addition, vehicles and road users will affect the extraction of background features, which dramatically reduces the detection accuracy.

III. LiDAR Sensor and Its Data

Compared to high-channel LiDAR sensors, such as 64-channel or 128-channel LiDAR sensors, the low-channel LiDAR sensor is expected to drop significantly in price to approximately $100 if massive production is realized [10]. In the future, the low-channel LiDAR sensor as an infrastructure-based unit will have a broader application in the fields of micro-level and high-resolution traffic information collection to provide complementary BVR traffic data for connected AVs, traffic management and control, collision warding, and so on.

Therefore, in this article, we used a low-channel LiDAR sensor (Velodyne VLP-16) to extend its detection range by constructing the static background from its point cloud data.

The VLP-16 sensor's vertical field of view is $\pm 15°$, and its vertical resolution is $\omega_r = 2°$. The horizontal field of view is 360°, and the horizontal resolution is 0.1°–0.4° according to the rotation speed. If the rotating frequency is set as 10 Hz, the horizontal resolution $\alpha_r \approx 0.2°$. Then, the maximum point number N_f within a single frame can be calculated as

$$N_f = \frac{360° * N_c}{\alpha_r} \tag{1}$$

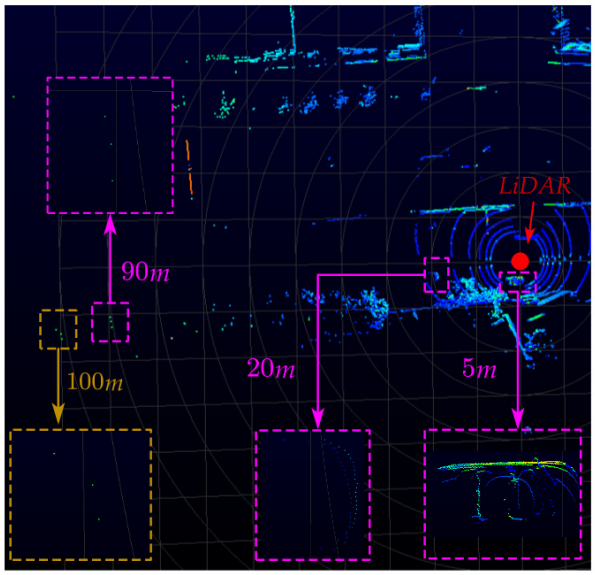


Fig. 1. Point number indicates a vehicle at a different distance within a VLP-16 LiDAR point cloud.

where N_c is the channel number of LiDAR and α_r is the LiDAR's detection horizontal resolution. For VLP-16 LiDAR, N_f is approximately 28 800 per frame.

The revolutions per minute (RPM) of LiDAR determine the horizontal resolution. There exist an azimuth range during LiDAR scanning. For VLP-16 LiDAR, the azimuth range is $[0°, 359.99°]$. Therefore, it has a deviation angle in the successive LiDAR point cloud frames, which is determined by the LiDAR mechanical structure and its built-in features. We defined the minimum deviation angle as $\alpha_{\min}$. For VLP-16 LiDAR, $\alpha_{\min} = 0.01°$. In the scanning area, there may exist maximum distinctive laser points N_{BP} as

$$N_{\mathrm{BP}} = \frac{360° * N_c}{\alpha_{\min}}. \tag{2}$$

For VLP-16 LiDAR, N_{BP} equals to 576 000, which is much larger than N_f.

Due to the mechanical structure of LiDAR and its built-in feature, all the laser beams are firing from the same original point with different angles. The density of LiDAR points decreases exponentially with increasing distance. Fig. 1 shows a typical point cloud from the VLP-16 LiDAR sensor for a traffic scenario where the vehicle is in the purple rectangle. At 5, 20, and 90 m away, the vehicle point numbers are 1426, 53, and 3, respectively. It is difficult to distinguish the vehicle points from background points as the distance increases. For instance, LiDAR points within the yellow rectangle 100 m away are the background points, and it is difficult to distinguish the vehicle point 90 m away within the purple rectangle.

IV. Static Background Construction

A. Background Construction

There will be a simple and effective way to construct a static background using point cloud frames without vehicles and road users within the scanning range [26]. However,

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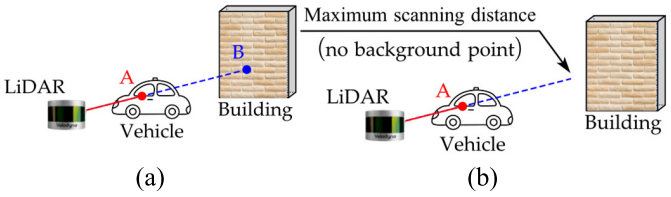


Fig. 2. Schematic of background objects occluded by vehicles. (a) Background object within the scanning range. (b) Background beyond the scanning range.

background object points will fluctuate over frames due to the vibration or swaying caused by the weather or LiDAR itself [7]. In addition, it is hard to find out the point cloud without vehicles and road users. Furthermore, background objects are often occluded by vehicles or road users in heavy traffic scenarios. Therefore, the vehicles, road users, and noise points should be removed when constructing the static background.

The background objects can be divided into in-range objects and out-range objects. The laser beam will return different laser points for in-range objects when a vehicle or road user passes. For out-range objects, the laser beam will return laser point when a vehicle or road user passing, else it will return nothing, as shown in Fig. 2.

As shown in Fig. 2(a), the building is within the LiDAR sensor scanning range. The LiDAR sensor will obtain point "A" when the vehicle is passing. Otherwise, the LiDAR sensor will get the point "B" of the building. In Fig. 2(b), the building is beyond the LiDAR sensor scanning range. The LiDAR sensor will obtain point "A" when the vehicle is passing. Otherwise, the sensor will return nothing.

A 2-D matrix $\boldsymbol{M_{BP}}$ was used to represent the background point cloud as

$$\begin{cases} \boldsymbol{M_{BP}} = \begin{bmatrix} d_{00} & \cdots & d_{0j} & \cdots & d_{0n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ d_{i0} & \cdots & d_{ij} & \cdots & d_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ d_{m0} & \cdots & d_{mj} & \cdots & d_{mn} \end{bmatrix} \\ \text{s.t.} \begin{cases} n = \dfrac{360^\circ}{\alpha_{\min}} \\ m = N_c \\ \omega_i = i * \omega_r \\ \alpha_j = j * \alpha_{\min} \end{cases} \end{cases} \tag{3}$$

where i is the number of rows; m is the maximum number of rows; N_c is the channel number of LiDAR; j is the number of columns; n is the maximum number of columns; $\alpha_{\min}$ equals 0.01°; ω_r means vertical angle resolution; ω_i means the ith vertical angle, which equals $i * \omega_r$; α_j means the jth horizontal angle, which equals $j * \alpha_{\min}$; and d_{ij} means the distance between the point and the LiDAR at the vertical angle ω_i and the horizontal angle α_j.

If points have the same vertical angle, the same horizontal angle, and form the same laser beam in the point cloud, the farthest distance point was defined as the background point as

$$d_{ij} = \begin{cases} 0; & D_{ij} \exists 0 \\ \max\left(D_{ij}\right); & \text{others} \end{cases} \tag{4}$$

where d_{ij} is the element of ith row and jth column in the matrix $\boldsymbol{M_{BP}}$, and D_{ij} is the set of distances of the points at the horizontal angle α_j and vertical angle ω_i in the point cloud for constructing the static background.

If an object is beyond the LiDAR scanning range, the laser will return $d_{ij} = 0$. Besides, there are still some adverse conditions that d_{ij} will return to 0. For example, the material absorbs the laser power, then d_{ij} will be turned to 0.

To identify the abnormal conditions, points in multiple frames with a distance equal to 0 were extracted and analyzed. The percentage of points P_{d0} at each rotation angle α_j which the distance is equal to 0 is calculated. The results are shown in Fig. 3.

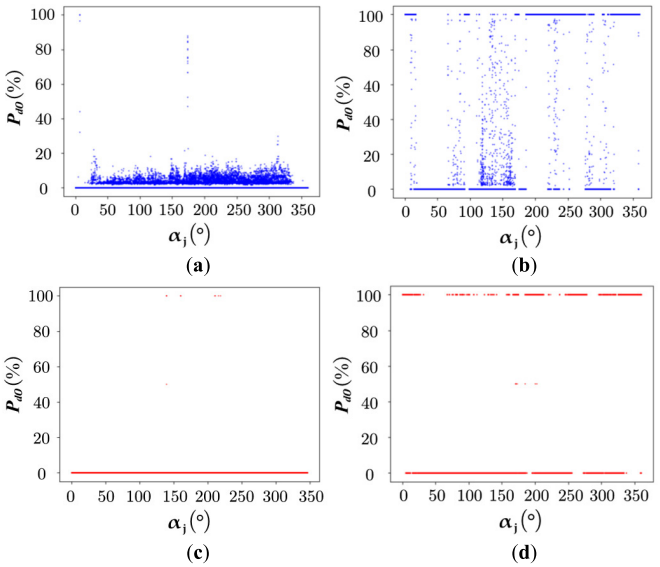


Fig. 3. Percentage of points P_{d0} at each rotation angle α_j which the distance is equal to 0. (a) In the heavy traffic scenario and $\omega_i = -15^\circ$. (b) In the heavy traffic scenario and $\omega_i = +15^\circ$. (c) In the scenario with few road users and $\omega_i = -15^\circ$. (d) In the scenario with few road users and $\omega_i = +15^\circ$.

In this article, the LiDAR is horizontally installed, and the mounting height is 1.8 m. According to the LiDAR sensor built-in feature, if $-15^\circ \le \omega_i < 0^\circ$, the laser beams will fire to the ground and some background targets, such as guardrails and low buildings. Then, the return distance value should be always more than 0. Therefore, in the scenario with few vehicles and road users, the value of P_{d0} will be stable at 0, as shown in Fig. 3(c). However, in the heavy traffic scenarios, there are many values of P_{d0} larger than 0, as shown in Fig. 3(a). Resulting in the loss of 0%–20% points and their distance is 0, as shown in the red rectangles in Fig. 4. We called this phenomenon as Error 1.

If $0^\circ \le \omega_i < +15^\circ$, the laser beams will fire to the sky, buildings, trees, and infrastructures. When firing to sky, the output value of LiDAR will be 0; otherwise, the value will be greater than 0. Therefore, in the scenario with few vehicles and road users, the value of P_{d0} will stable enough,

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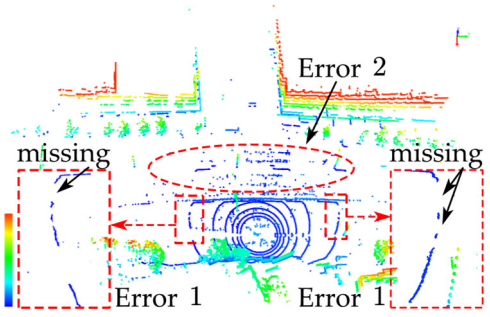


Fig. 4. Typical error in static background construction.

either 0% or 100%, as shown in Fig. 3(d). However, in the heavy traffic scenario, there are many values of P_{d0} between 0% and 100%, as shown in Fig. 3(b). This phenomenon is caused by the passing vehicles or road users. And the passing vehicle or road user points will be identified as background points, as shown in the red ovals in Fig. 4. We called this phenomenon as Error 2.

Errors 1 and 2 will reduce the reliability and accuracy of the static background in background filtering. Therefore, it needs to be optimized to reduce Errors 1 and 2 as much as possible.

B. Static Background Optimization to Reduce Points Missing

A threshold p_{th} is set to optimize the static background. Therefore, (4) is rewritten as

$$d_{ij} = \begin{cases} 0; & p_{ij} > p_{\text{th}} \\ \max\left(D_{ij}\right); & \text{others} \end{cases} \tag{5}$$

where p_{ij} is the ratio with the detected distance equal to 0 at ω_i and α_j, and p_{th} is the threshold value for determining whether Error 1 occurs at this angle.

However, the threshold setting to solve Error 1 will affect the occurrence of Error 2, as shown in Fig. 5.

Fig. 5 shows that as the value of p_{th} increases, the background information becomes richer in the near-range. The missing background points decrease as p_{th} increases and then remain without a significant change, although p_{th} increases, as shown in the black rectangles of Fig. 5. When p_{th} increases from 0.1 to 0.5, the missing background points decrease. However, when p_{th} increases to 0.75, there is no remarkable change compared to $p_{\text{th}} = 0.5$.

As p_{th} increases, the noise points become denser, as shown in the red oval of Fig. 5. Therefore, the LiDAR needs to find the best p_{th} to keep the balance of reducing missing background points and increasing the noise points. In this article, p_{th} is set as 0.5 to obtain the maximum of reducing background points missing.

C. Static Background Optimization to Filter Noise Points

Noise points are found frequently in areas where vehicles and road users pass, such as lanes and sidewalks. These areas are defined as passage areas (noted as A_m) and could be identified easily to filter the noise points.

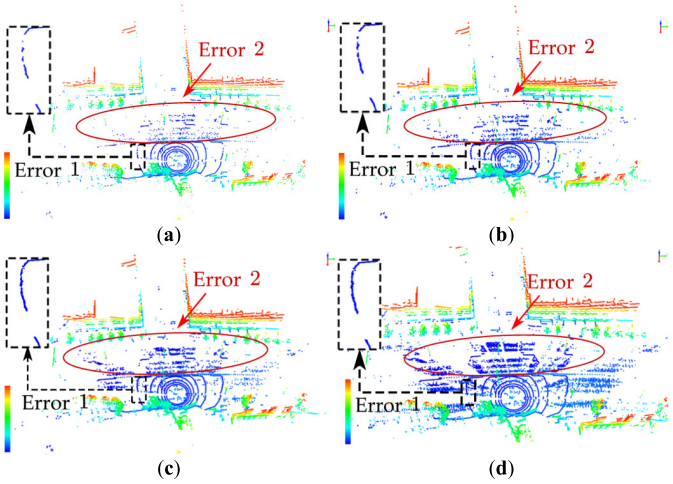


Fig. 5. Results of the static background construction by setting different p_{th}. (a) $p_{\text{th}} = 0.1$. (b) $p_{\text{th}} = 0.25$. (c) $p_{\text{th}} = 0.5$. (d) $p_{\text{th}} = 0.75$.

1) Passage Area Identification: According to the distance of points with and without vehicle or road user passing, it can be easy to identify whether the laser point P_t belongs to the vehicle or road user point

$$\text{IF}\left(d_{ij} - d_{ij}^t < d_{\text{th}}\right)\text{or}\left(d_{ij} = 0\right)\ \text{THEN}\ P_t \in Ps_1 \tag{6}$$

where d_{ij}^t and d_{ij} are the distances of points with and without the vehicle or road user passing in the horizontal angle α_j and vertical angle ω_i; Ps_1 is the point set of vehicles and road users; and d_{th} is the distance threshold. In this article, d_{th} was set as 0.5 m based on the experimental data that when $d_{\text{th}} \geq 0.5$ m, there was no significant difference in identifying the objects.

A density-based spatial clustering of applications with noise (DBSCAN) algorithm was used to identify the vehicles and road user points and generated Ps_1. There are two important parameters in DBSCAN: minimum number of points (MinPts) and epsilon (ε). In this step, we set Minpts $= 5$ and $\varepsilon = 1$ m to cluster vehicles and road users. The cluster results can be seen in Fig. 6.

It can be seen from Fig. 6 that there still exists some background objects in addition to vehicles and road users in the cluster results. This is due to the current background point cloud contains error points (Errors 1 and 2) and needs to be further optimized. In addition, to remove background objects and obtain an accurate passing area, we used 3-D boxes to surround the cluster objects. The same location of the 3-D box in the static background was extracted, and the point number in the 3-D box was used to compare. If the point number of 3-D box in the cluster is more than that in the static background, then the 3-D box was identified as a vehicle or road user.

The aggregation of trajectories of vehicles and road users within the multiframe of point clouds can be identified as the passage areas. Then, by mapping 3-D points of vehicles and road users into a 2-D plane, the passage areas in the background can be identified, as shown in Fig. 7.

Fig. 7(a) shows that the passage areas in the 2-D plane are discontinuous due to occlusion. A morphological processing

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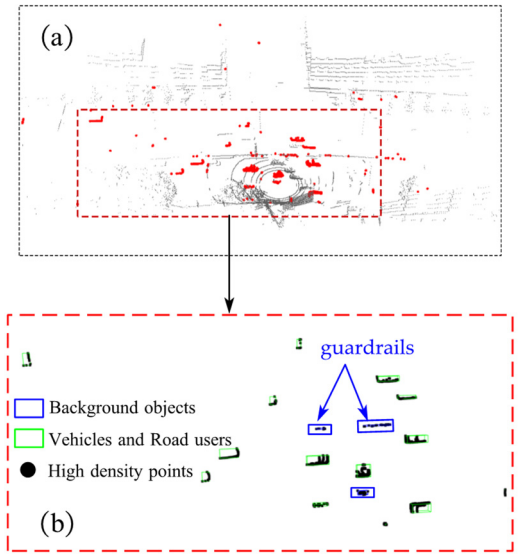


Fig. 6. Cluster results. (a) Cluster results in the background environment. (b) Cluster results and the classification of the results.

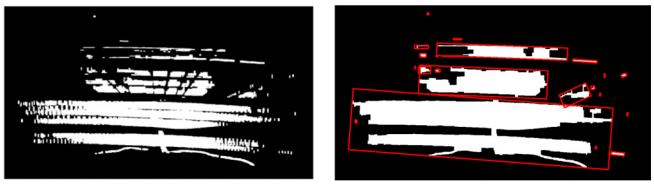


Fig. 7. Result of the passage area identification. (a) 3-D points of vehicles and road users transform into 2-D images. (b) Result of morphological processing for the 2-D images of vehicles and road users.

method based on reference [36] was used to improve the identification result, as shown in Fig. 7(b). The passage area in the red rectangles of Fig. 7(b) is defined as A_m in this article.

2) Noise Points Filtered From Passage Area: In the passage area A_m, there are mixed noise points and a large number of ground points. To filter noise points from A_m, the DBSCAN algorithm was used to extract ground points. Then, the noise points can be removed from A_m, as shown in the black rectangles in Fig. 8.

However, some noise points still remain in the background, which are not in the passage area A_m, as shown in the red ellipses of Fig. 8(b). Therefore, the static background needs to be further enhanced.

3) Noise Points Filtered From No Passage Area: The density of the LiDAR point cloud is variable and decreases exponentially as the distance increases. The distribution of noise points also follows the same characteristic. These noise points are not only far away from the background object, but also far away from each other. Therefore, these noise points can be considered as outliers in the background point cloud. The outlier removal algorithm in Open3d [37] can remove the outliers for this distribution characteristic. To identify and remove the noise points, the method needs two parameters, $nb_{\text{neighbors}}$ and std_{ratio}. In this article, $nb_{\text{neighbors}} = 20$ is used to specify the number of neighbors for a point and $std_{\text{ratio}} = 5$ is

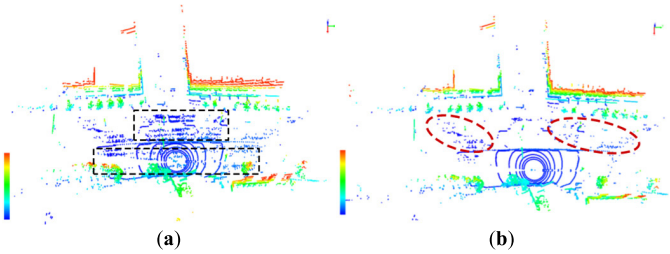


Fig. 8. Before and after noise point removal from A_m. (a) Background with noise points is in the red oval. (b) Background with noise points removed from A_m.

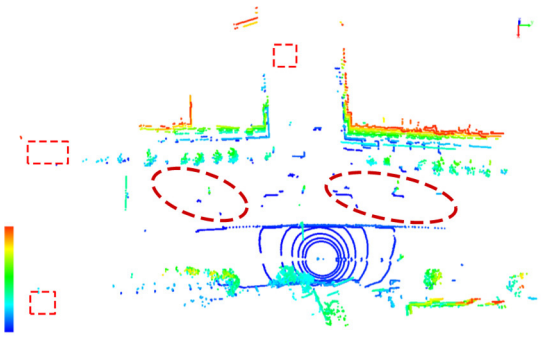


Fig. 9. Results after removing points by Open3d.

the standard deviation of the point distance. The results are shown in Fig. 9.

The noise points in the red ellipses in Fig. 9 were removed successfully. However, some distant background object points were also removed because their density is too sparse, as shown by the red rectangles in Fig. 9.

Although background points are indistinguishable from sparse vehicle or road user points in density, they differ significantly from the vehicle or road user point. Even though only one point is scanned for background objects, the points of background objects will always appear in the same position in the point cloud frames.

Therefore, we obtained all points at the corresponding positions of the removed outliers from the point cloud used for background construction. Furthermore, to eliminate the influence of different vehicles or road users in the same position at different times, we randomly select N_3 frames of point clouds for each labeling process. The smaller N_3 is, the more road user points will be labeled. This article sets $N_3 = 95\%N_s$, where N_s is the total number of frames for the construction. Then, road user points are labeled repeatedly until the number of vehicles or road users remains unchanged, as shown in Fig. 10.

Fig. 10 shows that after 50 iterations, the number of labeled points is unchanged. Finally, the unlabeled points in far-range areas were reserved for constructing the static background.

V. Extending Detection Range of Roadside LiDAR

A. Background Filtering

With the static background constructed in Section IV, a 2-D matrix $\boldsymbol{M_{BP}}$ for the static background can be obtained. Then, a ground plane can be formed by using the background points.

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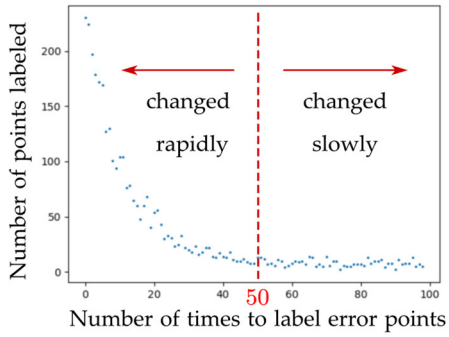


Fig. 10. Changing trend between the number of labeled points and the time of repeat labeling.

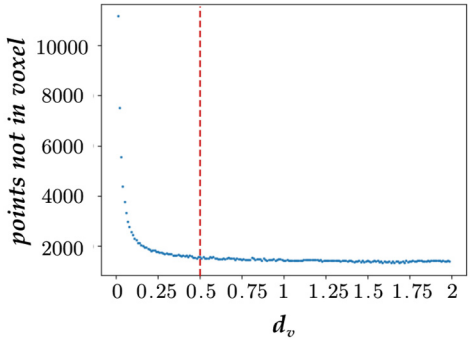


Fig. 11. Relationship between d_v and the number of points excluded in the voxels.

When the position between a LiDAR and ground is stationary, the track on an ideal ground plane will be an ellipse or circle after a specific channel inside the LiDAR rotates for one circle. Therefore, a curve can be used to fit this track, and the points on the actual ground can be classified as normal ground points (NGPs) and abnormal ground points (AGPs) (curbs, holes, or obstacles). Then, we could find a plane with the least standard deviation of the distance from all NGPs as the ground plane (GP); more detail can be found in our previous work [38].

To filter a new raw point cloud, GP was used to filter the ground point first [38]. Then, the remaining points were compared to $\boldsymbol{M_{BP}}$ to filter the background points within the remaining points. The voxel method was used to filter the background point. The voxel size was set as d_v. The larger d_v is, the more points can be removed, but the background filtering precision will be lower. To find an appropriate value for d_v, a d_v value from 0 to 2 m was set in the experiments, as shown in Fig. 11.

As the value d_v increases to 0.5, the number of background points including the voxels increases enormously, after which the number of excluding points within the point cloud changes indistinctively.

For two-step background filtering, most of the background points and noise points can be filtered successfully, as shown in Fig. 12.

B. Detection

For the near-range, the DBSCAN algorithm is used to detect vehicles and road users. In the algorithm, the parameters of the

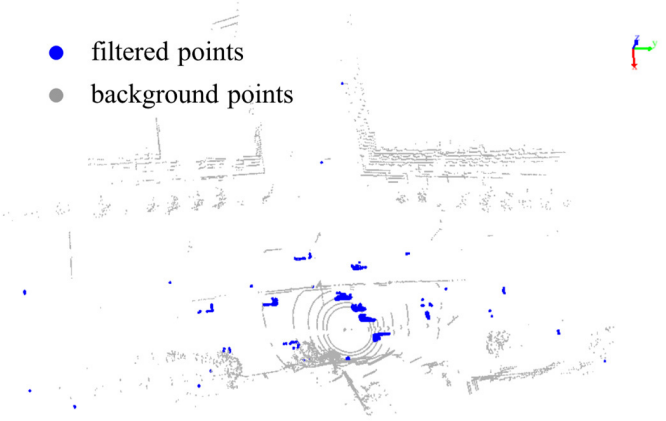


Fig. 12. Result of background filtering.

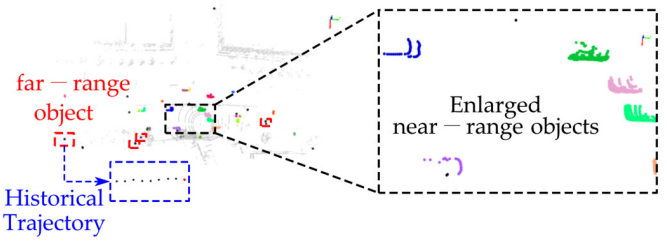


Fig. 13. Points distribution in the near-range and far-range.

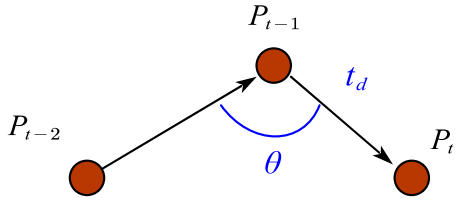


Fig. 14. Schematic of the object's displacement distance and direction.

minimum points Minpts and the radius ε were set as 3 and 1 m, respectively, according to the experiments in reference [13].

For the far-range, the point cloud is sparse and easily occluded by near objects. Only one or two points for a road user or vehicle will be detected by the LiDAR sensor, as the red rectangle in Fig. 13.

It is difficult to distinguish a single point of a vehicle or road user from a single-point cloud frame. Based on the trajectory tracking algorithm [14], [39], it can be found that even one point of a vehicle or road user is detected by the LiDAR sensor in the far-range within a single-point cloud frame, its trajectory and movement characteristics can be extracted by the multi-frame of point clouds data.

A moving object has two properties: distance and direction. Distance means the Euclidean distance of reference points of the same object in the adjacent point cloud frames. The direction needs to be determined by three reference points, as shown in Fig. 14.

As shown in Fig. 14, P_{t-2}, P_{t-1}, and P_t are the reference points of the same object in three adjacent point cloud frames. t_d is the displacement distance of the object in the adjacent point cloud frames. θ is the trajectory angle in the

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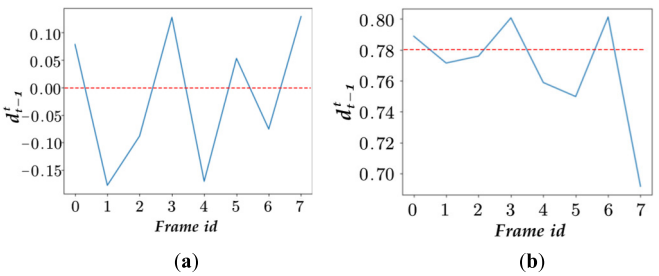


Fig. 15. Displacement distance of a noise point and a vehicle or road use point in adjacent point cloud frames. (a) Displacement distance of the noise point in adjacent point cloud frames. (b) Displacement distance of vehicle or road use points in adjacent point cloud frames.

three adjacent point cloud frames. In general, θ is close to 180° in a straight lane or sidewalk for vehicles or road users.

For noise points, the displacement distance and direction of the noise point are random and always bounce back and forth. The displacement distance and direction of the single point in the adjacent point cloud frames is described as

$$d^t_{t-1} = \begin{cases} t_d, & \theta > 90^\circ \\ -1 * t_d, & \text{others} \end{cases} \tag{7}$$

where d^t_{t-1} means a point from the $t-1$ point cloud frame to the t point cloud frame. $d^t_{t-1} > 0$ means that the current motion direction of the point is successive to the previous motion direction. Fig. 15 shows an example of the displacement distance of noise and a vehicle.

The displacement distance value of the noise point frequently fluctuates around approximately 0. However, the displacement distance of a vehicle or road user is always larger than 0 to a threshold value of vehicles or road users. From the perspective of signal processing, the displacement distance of the noise point has a larger alternating current (AC) component. Meanwhile, the displacement distance of the vehicle or road user point has a larger direct current (DC) component.

The FFT [40] is a common algorithm in signal processing that can transform the time domain signal into the frequency domain signal to analyze and identify the dc and ac components. Therefore, FFT is used to distinguish the noise point from vehicle or road user points, as shown in Fig. 16.

The value of the frequency equal to 0 is the amplitude of the dc component, denoted as A_{DC}, and the value of the frequency greater than 0 is the amplitude of the ac component with different frequencies, denoted as A_{AC}. The A_{DC} and A_{AC} between noise points and vehicle or road user points are quite different. The value of A_{DC} of noise is very small, near zero, while A_{AC} is larger. The opposite is true for vehicles and road users. Therefore, the condition for determining vehicle or road users can be

$$A_{\mathrm{DC}} > \max(A_{\mathrm{AC}}) \text{ and } A_{\mathrm{DC}} > A_{\mathrm{th}} \tag{8}$$

where A_{th} is the threshold of the minimum amount of movement of a vehicle or road user. For VLP-16 LiDAR, the rotation frequency is approximately 0.1 s, and the average speed of pedestrians is approximately 1 m/s [41]. Therefore, this article sets $A_{\mathrm{th}} = 0.1 * 1 = 0.1$ m.

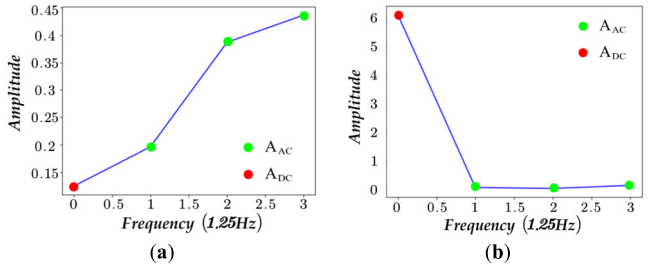


Fig. 16. FFT results of noise points and vehicle or road user points. (a) FFT result of noise points. (b) FFT result of vehicles or road user points.

TABLE I
RESULTS OF DIFFERENT n_f ON VEHICLE AND ROAD USER DETECTION

n_f	Average Precision (%)	Average recall (%)
2^2	92.04	80.81
2^3(selected)	**93.12**	**89.38**
2^4	90.01	85.52
2^5	87.42	78.91

Sampling number n_f is an important parameter of FFT, which is usually equal to 2^k, and k is a positive integer. Obviously, n_f is fixed, while far-range objects are easily obscured by nearby objects, which may result in incomplete trajectories, the scanning period of LiDAR is fast, about 0.1 s. During the sampling period, moving pedestrians and vehicles can move at approximately uniform motion. Therefore, we used linear interpolation to the incomplete trajectory. In addition, the value of n_f will affect the distinguishing result. Therefore, n_f is selected by the enumeration method using the AP and AR to decide the best value. The results are shown in Table I.

The best detection results of road users is at $n_f = 2^3$. When $n_f = 2^2$, the poor performance is caused by less trajectory information. While $n_f = 2^4$ or 2^5, although it contains abundant trajectory information, it is easy to recognize the trajectories of multiple vehicles or road users within a too large period as one trajectory, resulting in the reduction of final results.

VI. CASE STUDY

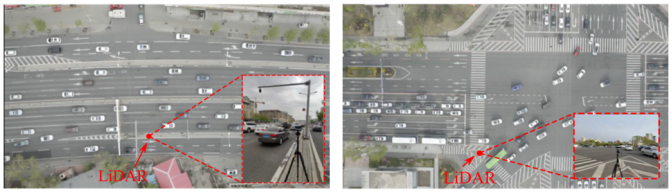
The data used in this article were obtained from the Velodyne VLP-16 LiDAR sensor (10 Hz). The sensor was installed on the roadside of the Yatai expressway and Gongnong signalized intersection, where the traffic flow is always heavy, as shown in Fig. 17.

On the roadside of the Yatai expressway, 15 000 frames of point cloud data were collected during the peak hours. On the roadside of the Gongnong signalized intersection, 9000 frames of point cloud data were collected during the peak hours.

A. Evaluation of the Static Background Construction

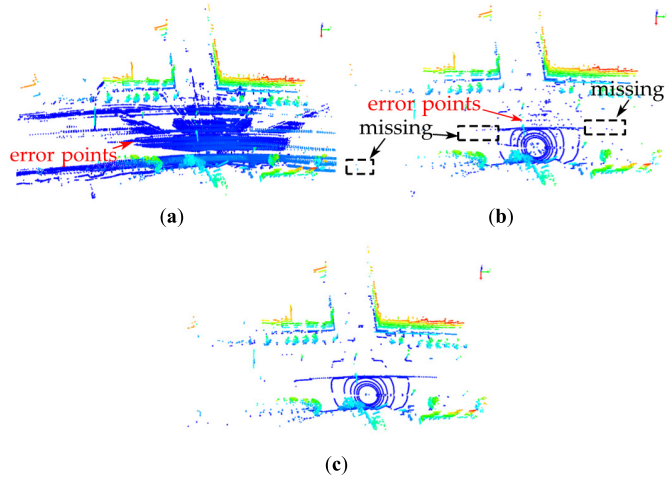
To evaluate the static background construction, Wu's background construction method [9] and Zhang's background construction method [13] were used to be compared, as shown in Fig. 18.

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(a) (b)

Fig. 17. Snapshot of the location of LiDAR point cloud data collection. (a) Yatai expressway in Changchun, China (Location 1, 43.8558°N, longitude 125.3406°W). (b) Gongnong signalized intersection in Changchun, China (Location 2, 43.8552°N, 125.3318°W).



(a) (b)

(c)

Fig. 18. Result of background construction by different methods. (a) Wu's background construction method. (b) Zhang's background construction method. (c) Our method.

TABLE II
NUMBER OF FRAMES FOR BACKGROUND CONSTRUCTION

Installation Sites	Methods	The number of frames
Yatai Expressway	Wu's work [9]	3000
	Zhang's work [13]	856
	Our work	**315**
Gongnong Signalized intersection	Wu's work [9]	3000
	Zhang's work [13]	1869
	Our work	**1200**

For Wu's work [9], the number of aggregated data frames for background identification was 3000. Fig. 18(a) shows that the information of the construction background is rich, but there are too many vehicle or road user points contained in the background points. For Zhang's work [13], the value of the convergence condition threshold δ was set to 0.19 m, and many background points were lost in the construction background, as shown in the black rectangles of Fig. 18(b). By comparison, the static background construction by the proposed method is minimally interfered with by vehicles or road users, as shown in Fig. 18(c).

In addition, the number of frames needed in background construction is another comparable index, as shown in Table II.

At the Yatai expressway, the traffic volume is large and continuous. A total of 315 frames of the point cloud are needed in constructing a static background to cover the vertical angle and horizontal angle of the LiDAR sensor and rich background information. In the Gongnong signalized intersection, the traffic volume is heavy but dispersed by the traffic signalized, and 1200 frames of the point cloud are required to ensure that the vehicles in each lane move at least once. However, fewer frames were needed in background construction than in Wu's and Zhang's works.

TABLE III
DETECTION RANGE WITH DIFFERENT METHODS

Methods	Max distance of detection (m)		
	Vehicle	Cyclist	Pedestrian
Slice-based [27]	30	25	22
Zhang's work [13]	**100**	45	43
Our work	**100**	**100**	**85**

TABLE IV
AVERAGE PRECISION AND RECALL OF DETECTION

Installation Sites	Methods	Average precision (%)	Average Recall (%)
Yatai Expressway	Slice-based [27]	90.03	32.18
	Zhang's work [13]	63.63	86.27
	Our work	**94.79**	**90.62**
Gongnong Signalized intersection	Slice-based [27]	81.16	29.25
	Zhang's work [13]	79.81	74.64
	Our work	**95.71**	**84.81**

B. Evaluations of Vehicle and Road User Detection

Furthermore, we compared the detection range and its AP and recall with the slice-based method [27] and Zhang's work [13]. The results are shown in Tables III and IV.

Pedestrian detection is the most difficult issue for far-range detection due to its point number and motion characteristics. Therefore, the detection range for pedestrians is always minimal regardless of the method used in detection. However, the detection range of our proposed method can be effective up to 85 m. In addition, the detection range for vehicles and cyclists can be up to 100 m, which is equal to the claimed detection range of the VLP-16 LiDAR sensor.

Furthermore, to measure the effectiveness of the detection, 1000 frames of point clouds were tested in the experiment, as shown in Figs. 19 and 20.

In Table IV, the AP and recall of our proposed method are higher. For the slice-based method [27], the value of the AP is much larger than the AR value because the detection range of the slice-based method [27] is very small. Too many vehicles and road users in the far-range are not detected, as shown in the red ellipses of Figs. 19(b) and 20(b).

For Zhang's work [13], the AR is lower than our work, which is due to the poor performance in pedestrian detection in the far-range. Meanwhile, some static objects are recognized as vehicles or road users, as shown in the black rectangles of Figs. 19(c) and 20(c).

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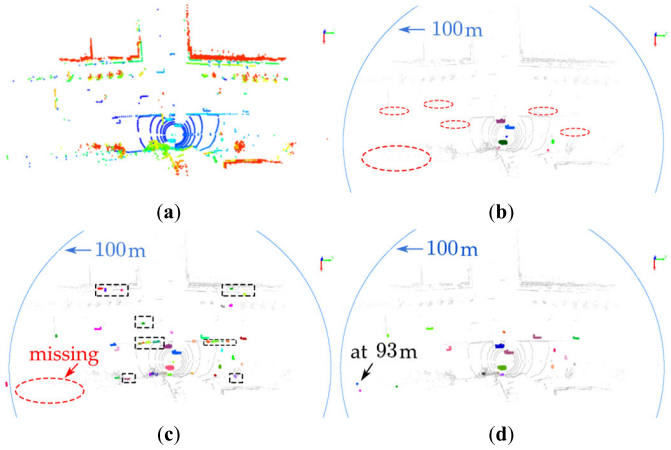


Fig. 19. Detection results by different methods. (a) Raw roadside point cloud data of the Yatai expressway. (b) Detection result by the slice-based method. (c) Detection result by Zhang's method. (d) Detection result by our proposed method.

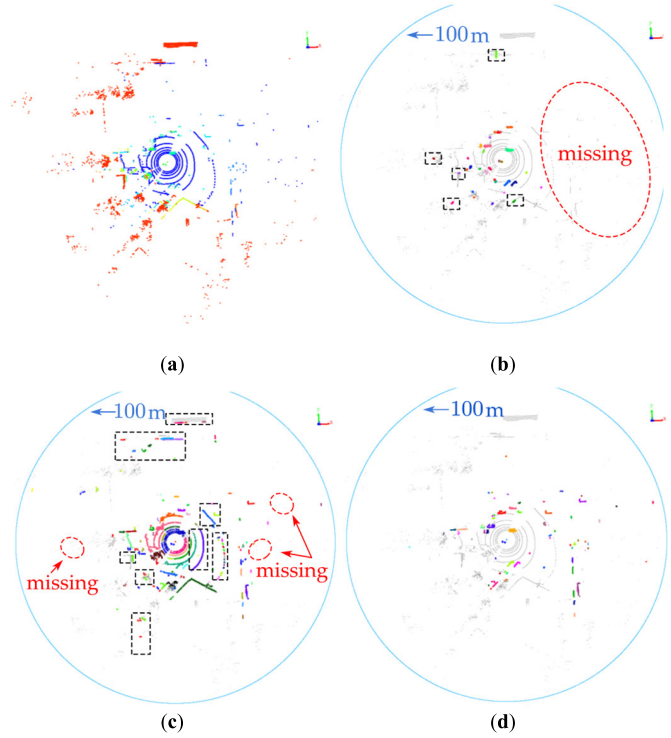


Fig. 20. Detection results by different methods. (a) Raw roadside point cloud data of the Gongnong signalized intersection. (b) Detection result is by the slice-based method. (c) Detection result by Zhang's method. (d) Detection result by our proposed method.

Figs. 19(d) and 20(d) show the results of our work, and more detailed information can be found in https://youtu.be/r-bvx_TjEwQ, which is a video that outperforms our proposed method of this article.

C. Evaluations of Method Complexity

This section compares the proposed method with the other two background-construction-based methods (Wu's work [9] and Zhang's work [13]) in terms of memory-space complexity and time complexity, as shown in Table V. These methods are tested under the same platform, using Python 3.8 programming language in 64-bit Windows 10, Intel Core i7, CPU 2.7 GHz (single thread), and 8-GB RAM.

TABLE V
COMPARISONS OF METHOD COMPLEXITY

Methods	Storage usage of background (MB)	Time of Background construction (min)
Wu's work [9]	259.2	25
Zhang's work [13]	**1.7**	**7**
Our work	**1.7**	310

For memory-space complexity: This article compared the memory space usage of the background point cloud generated in the methods. It takes major memory space in the process of real-time detection [42]. The size of our work, Zhang's work [13], and Wu's work [9] are 1.7, 1.7, and 259.2 MB, respectively. It can be seen that the size of the background point cloud in our work and Zhang's work is much smaller than that in Wu's work.

For time complexity: The time consumption of background construction progress of our work, Wu's work [9], and Zhang's work [13] are 310, 25, and 7 min, respectively. It can be seen that the time complexity of the background construction of the proposed method is much higher than that of the other two methods, as the proposed construction method spent more steps to construct the static background more comprehensively and integrally. Although the construction progress is time-consuming, it helps improve the detection precision, recall, and pedestrian detection range by 28.12%, 4.78%, 45 m, and 31.16%, 4.35%, 42 m, respectively, comparing with Wu's work [9] and Zhang's work [13]. In addition, the static background construction is performed offline before real-time traffic objects detection with online point cloud data.

VII. CONCLUSIONS AND DISCUSSION

In this article, we presented a method for extending the detection range for roadside LiDAR. First, a static background was constructed as a funnel to filter the background point. In the static background construction, multiple frames of point cloud data were used to cover the most horizontal angle and vertical angle of the LiDAR sensor and obtain background information ultimately. Furthermore, the static background was optimized to reduce the missing far-range background points and the occurrence of noise points. Compared with other methods, the proposed method in this article can construct a more accurate static background with less point cloud data.

Then, the vehicles and road uses were detected. For the near-range, the vehicles and road users were identified by the DBSCAN algorithm. For the far-range, the trajectory of objects is tracked based on their moving direction and distance, and then an FFT algorithm is used to identify the vehicle and road user points. The average detection range of vehicles and cyclists can be extended to 100 m. The average detection range of pedestrians can be extended to 85 m. Furthermore, the AP and recall of the proposed method are 95.71% and 90.62%,

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respectively. Compared with previous studies, the proposed method performs better in the detection range, AP, and recall with low-channel roadside LiDAR. In addition, more accurate and further detection will increase traffic safety by perceiving unexpected traffic incidents earlier, allowing more time to make decisions and implement these decisions.

However, the present study and its methodological approach are not free from limitations.

1) Vehicles and road users that are blocked by nearby objects could not be detected directly.
2) Far-range vehicles and road users were detected based on their motion characteristics. If the vehicles or road users are static, the vehicles or road users will not be identified.
3) The process of static background construction is time-consuming, and it usually takes several hours when using the Python 3.8 programming language.

Furthermore, there are still some topics to be further studied. Further research work includes the following.

1) The trajectories of vehicles or road users are often interrupted by blocks or missing data. Their trajectories need to be repaired and predicted to fill the missing or block.
2) Vehicles and road users should be identified and classified in more detail, improving the detection precision with their motion characteristics.
3) The proposed method needs to be performed using other 16-channel or more than 16-channel LiDAR sensors to compare.

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